

ELECTROSTATIC CYCLOTRON

PROBLEM STATEMENT

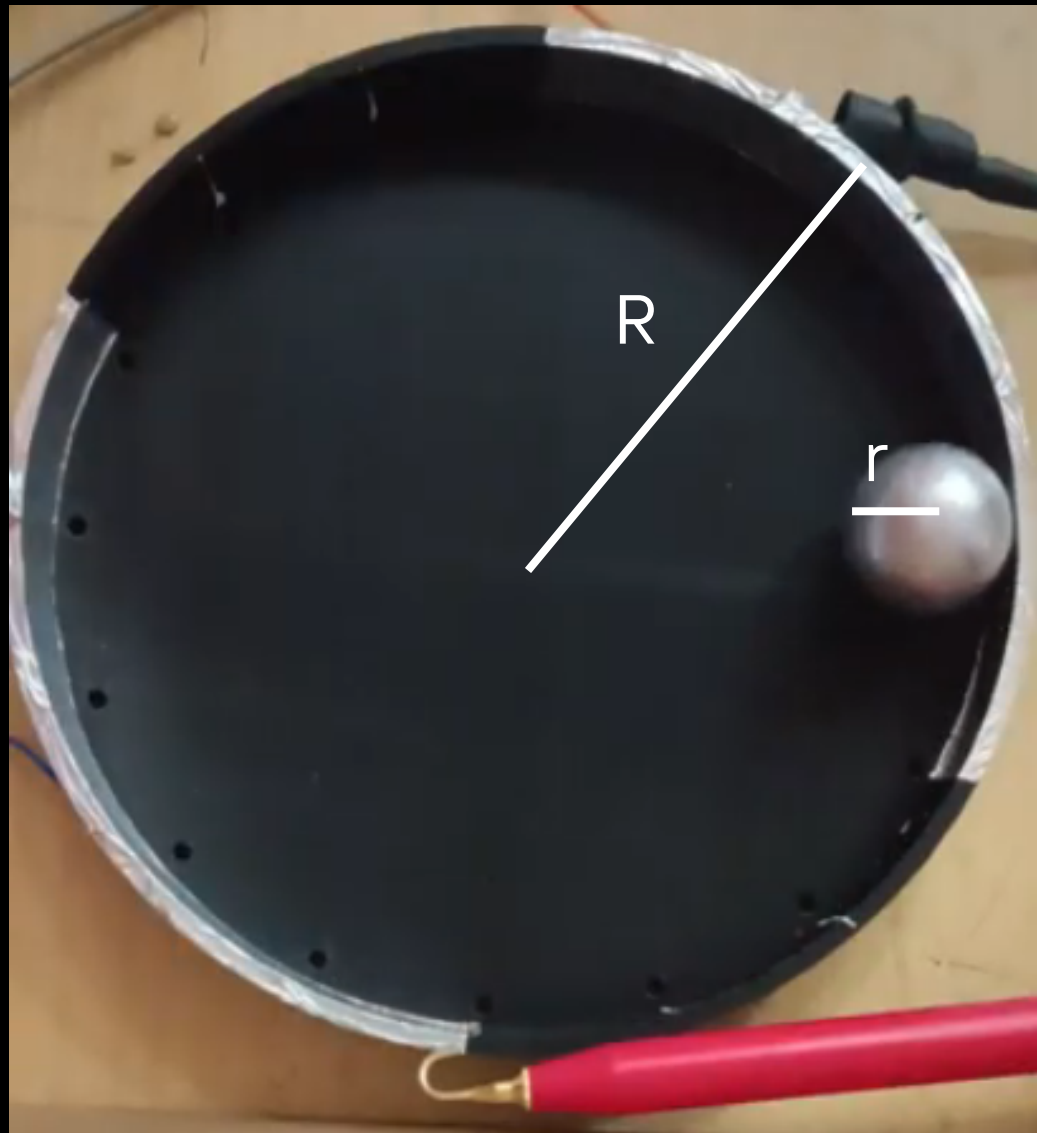


Using a DC voltage source and a plastic ball covered in aluminum foil, build the setup shown in the video. Determine under which conditions the setup can work and maximize the ball's speed.

THEORY AND SIMULATIONS

Page 03

Setup and Working Principle



The system consists of a lightweight conducting ball, coated with aluminum foil, placed between curved electrode strips arranged along a circular path and connected to a DC voltage source.

The applied potential creates an electric field in the region, enabling electrostatic interaction between the ball and the strips.

When the ball comes into contact with a strip, it acquires charge and experiences a repulsive electrostatic force.

This causes the ball to accelerate away from the strip and move toward the next electrode, where another charge transfer occurs.

As this process repeats, the system sustains motion through successive charge–repulsion cycles.

Assumptions

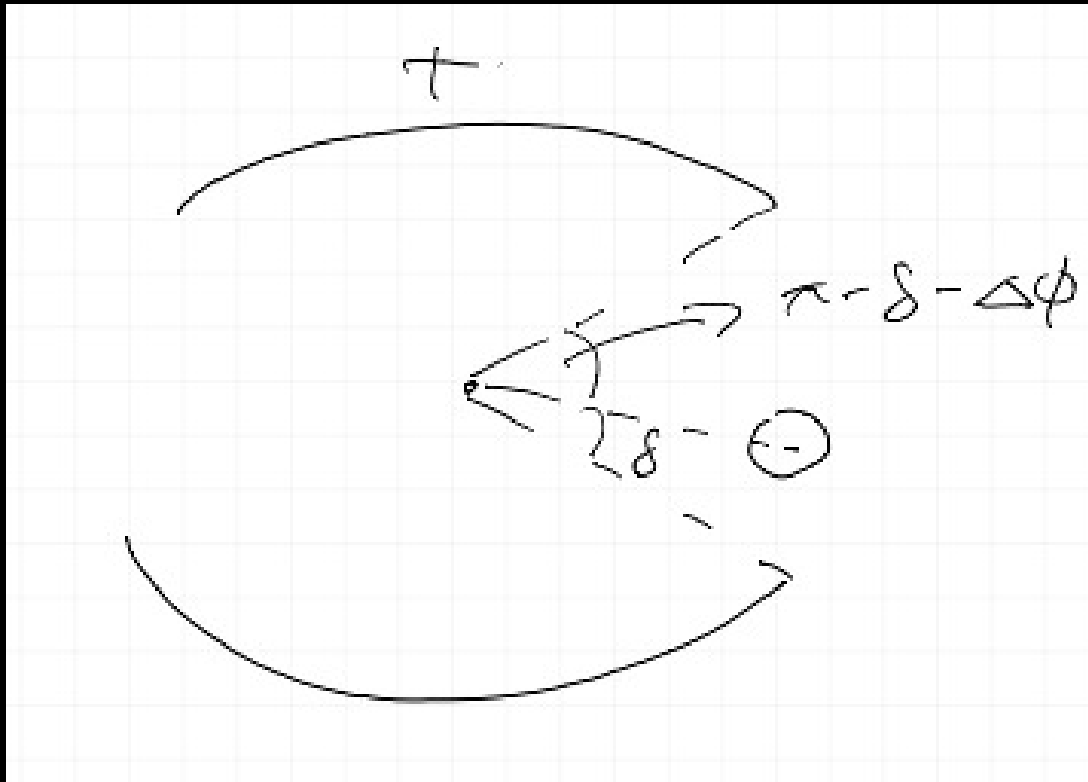
- Charge distribution on the ball and electrode strips is assumed uniform in the non-contact regime.
- Only the nearest electrode strips are considered, as distant strips' effects are weaker and partially cancel out due to symmetry.
- Analysis focuses solely on electrostatic interactions when the ball is not touching the strips.
- Dissipative effects like friction, air drag, and charge leakage are excluded from the model.

Geometry of the System

- Electrode strips are arranged on a circle of radius R .
- Ball center moves on a circle with radius $R-r$.
- Ball's angular position ϕ defines its location relative to strips.
- Distance between ball center and strip is

$$c(\phi) = R^2 + (R - r)^2 - 2R(R - r) \cos(\phi)$$

Force Model



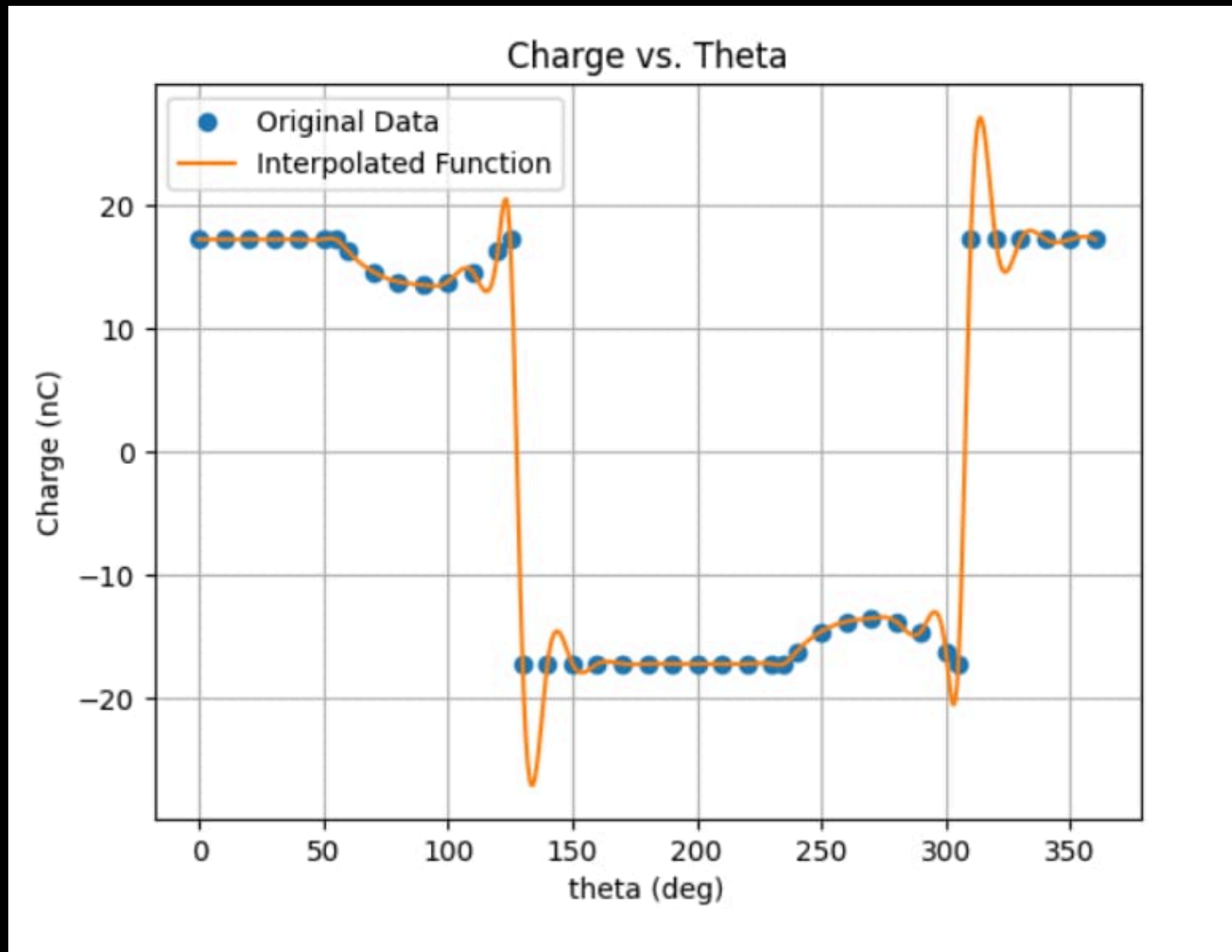
- The electrostatic force is calculated by integrating the Coulomb interaction between the charged ball and the charged strip over its finite extent.
- This results in a position-dependent tangential force acting on the ball.

$$F = -\frac{Q \sigma R}{8\pi\epsilon_0(R - r)} [\xi(\pi - \delta - \Delta\phi) + \xi(\delta)]$$

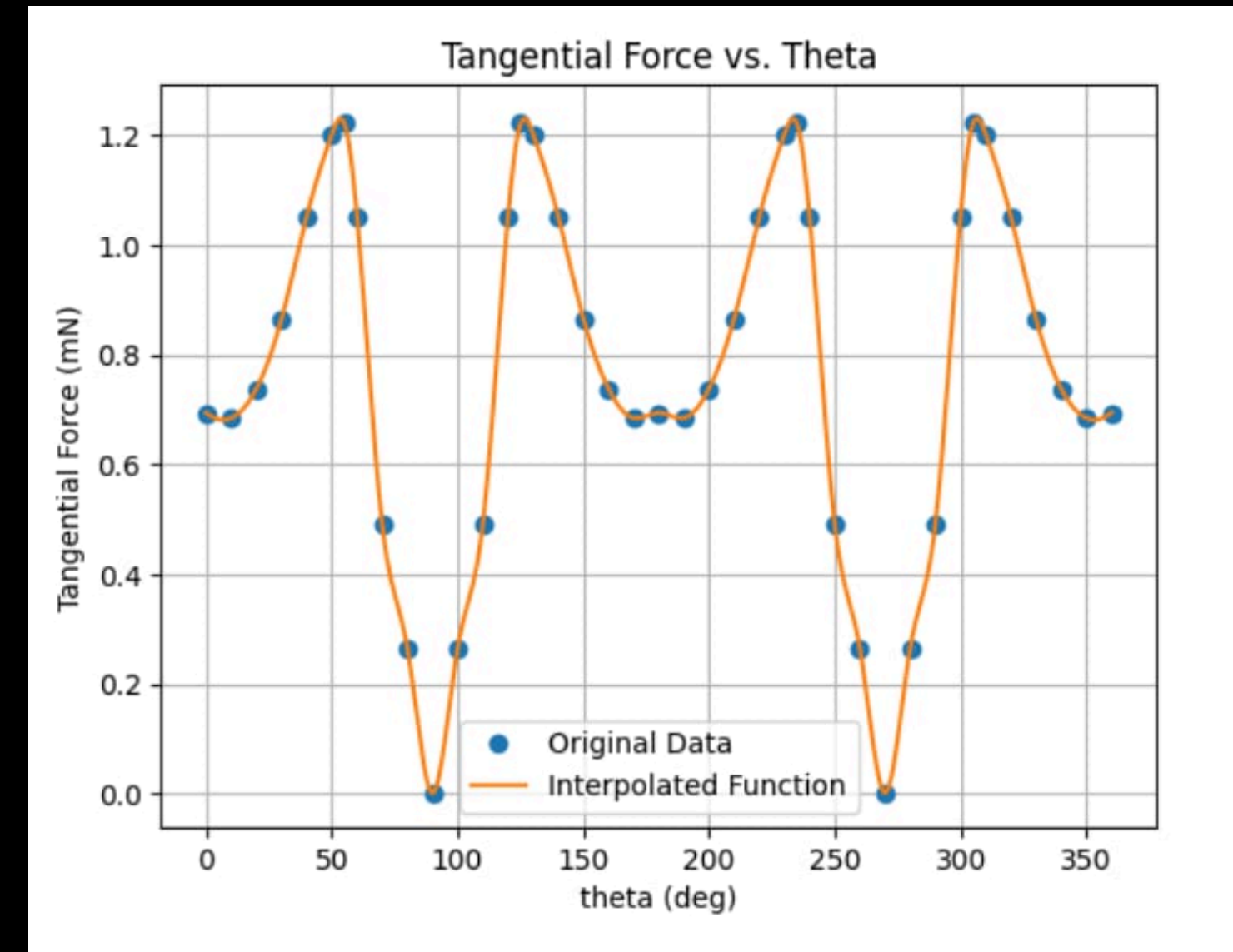
$$\xi(\phi) = \ln \left[\left(\frac{\sqrt{c(\phi + \Delta\phi) + \ell^2(1 - p)^2} - \ell(1 - p)}{\sqrt{c(\phi + \Delta\phi) + \ell^2(1 - p)^2} + \ell(1 - p)} \right) \cdot \left(\frac{\sqrt{c(\phi + \Delta\phi) + \ell^2 p^2} - \ell p}{\sqrt{c(\phi + \Delta\phi) + \ell^2 p^2} + \ell p} \right) \cdot \left(\frac{\sqrt{c(\phi) + \ell^2(1 - p)^2} - \ell(1 - p)}{\sqrt{c(\phi) + \ell^2(1 - p)^2} + \ell(1 - p)} \right)^{-1} \cdot \left(\frac{\sqrt{c(\phi) + \ell^2 p^2} - \ell p}{\sqrt{c(\phi) + \ell^2 p^2} + \ell p} \right)^{-1} \right]$$

COMSOL Simulation

Page 06



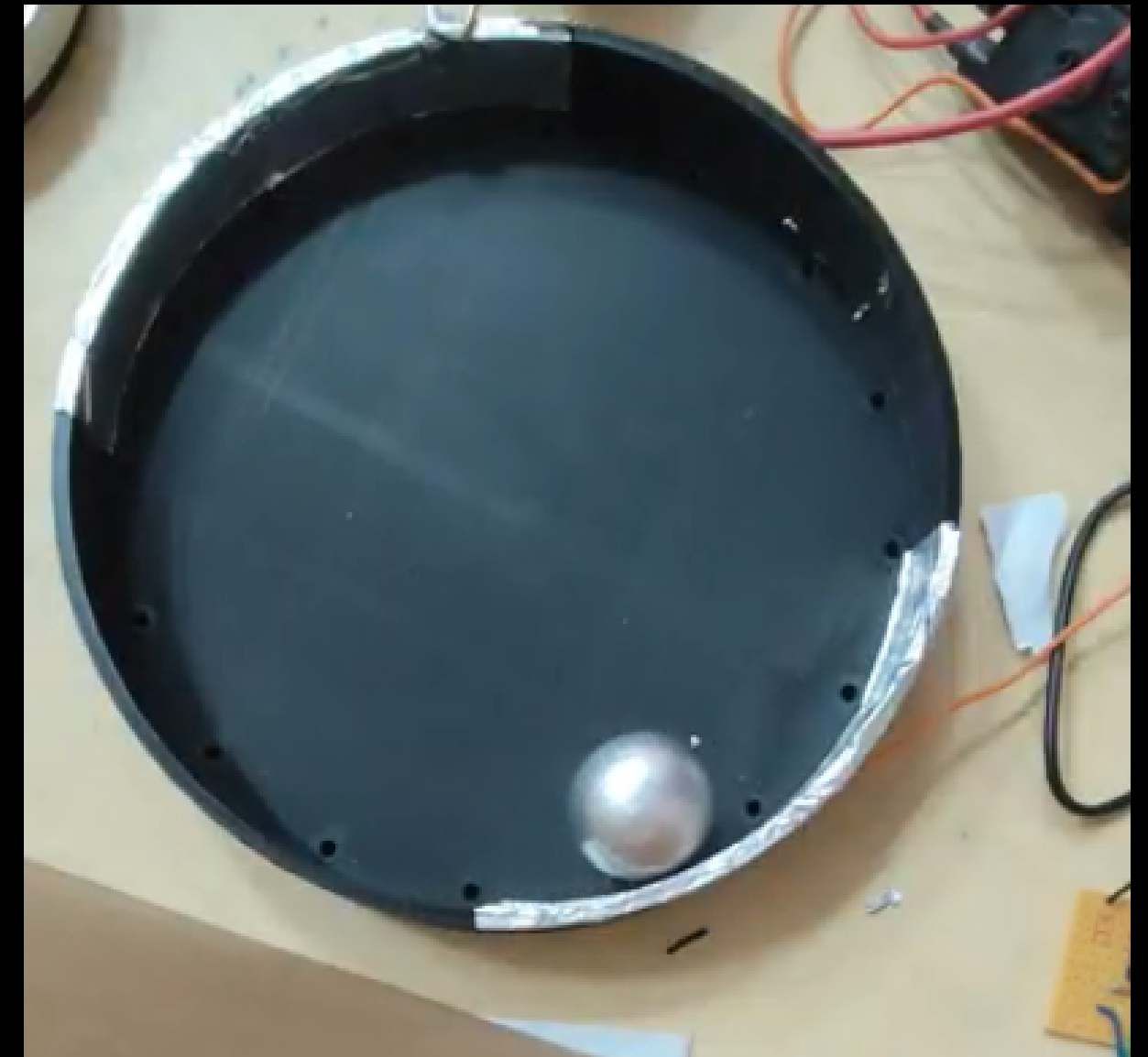
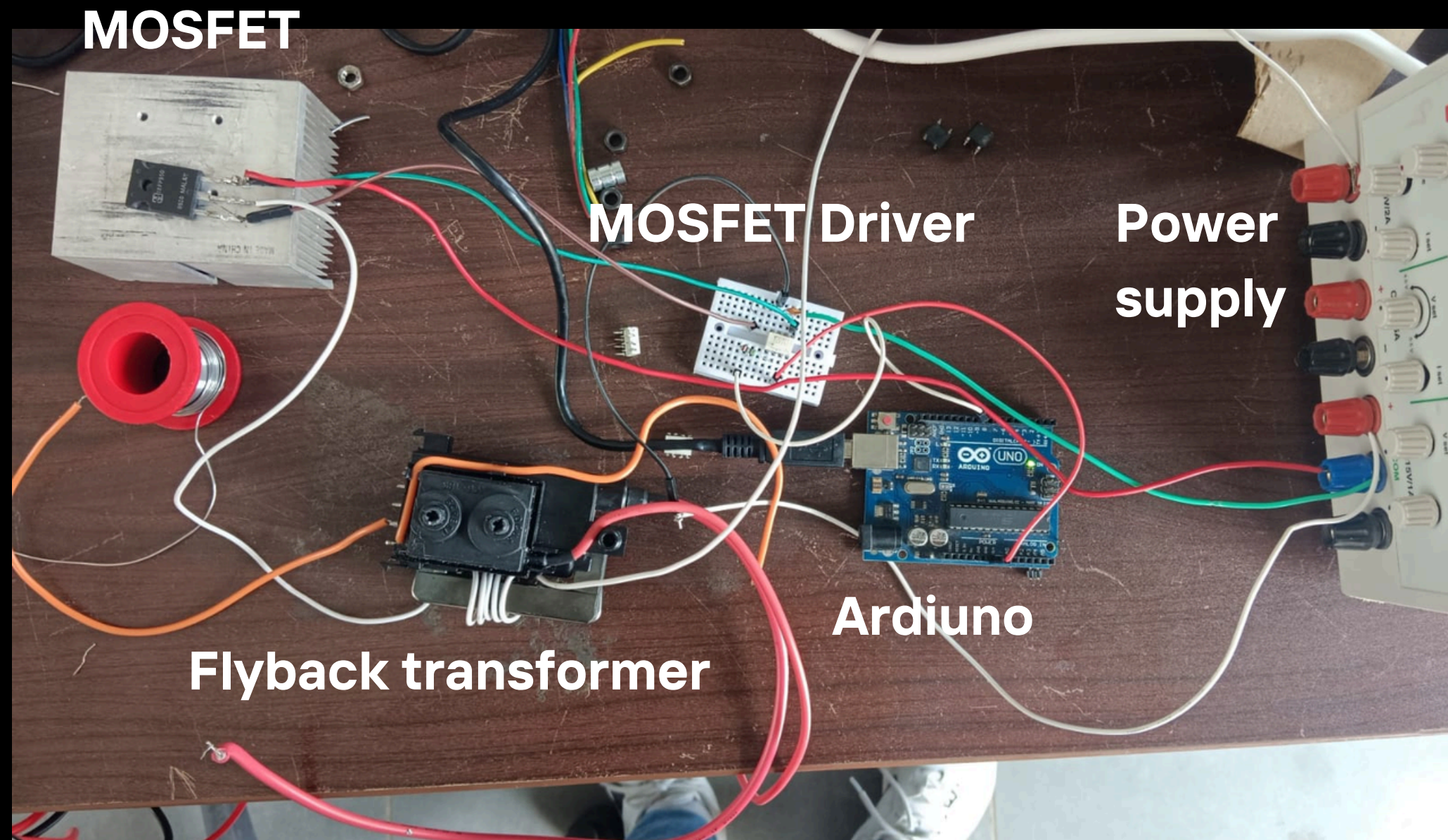
- Piecewise constant charge
- Sudden jumps at contact



- Periodic variation
- Peaks near strip edges

EXPERIMENTAL SETUP

Page 07



EXPERIMENTAL SETUP

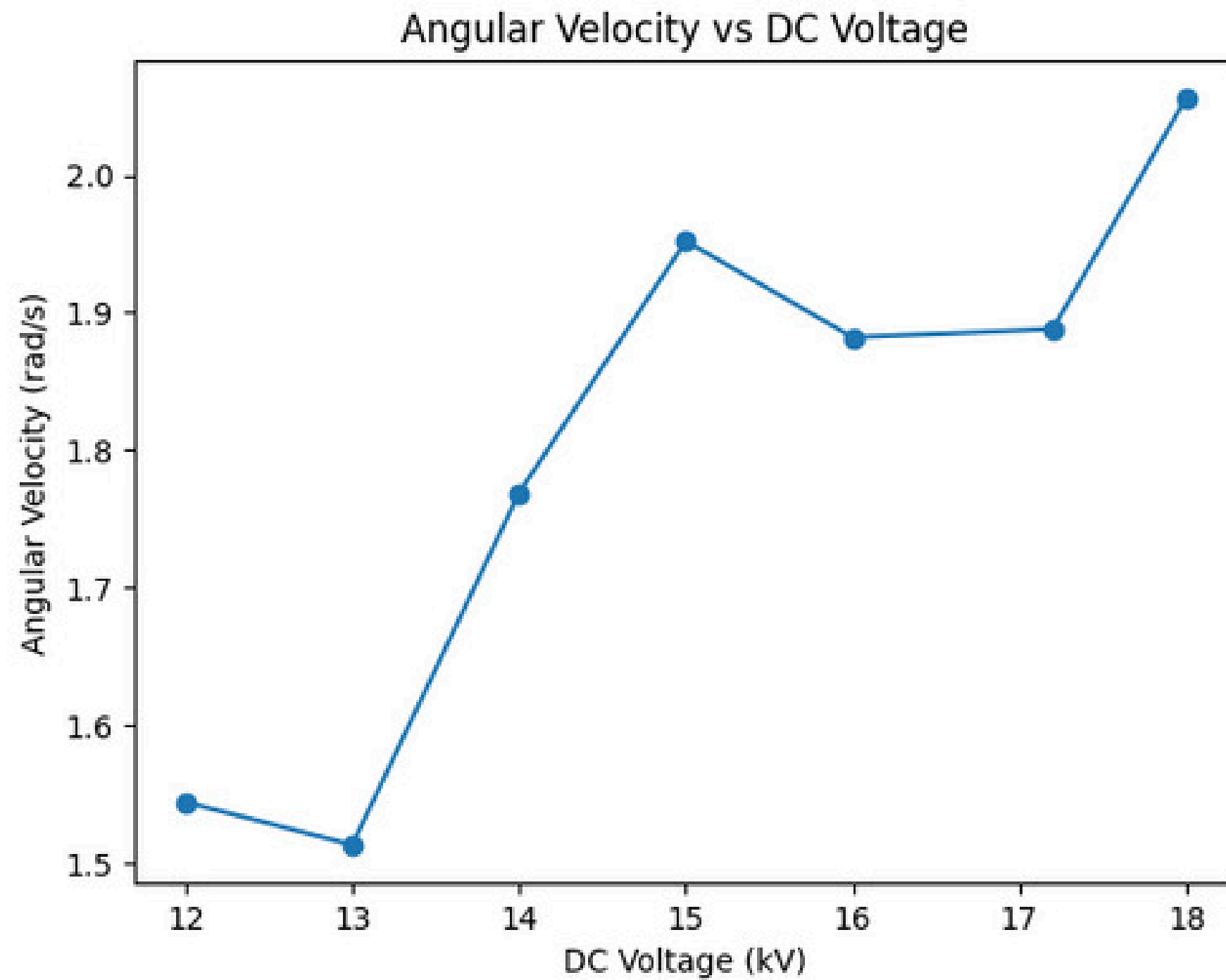
PARAMETERS

- DC voltage: 13 KV - 23 KV
- Radius of strip: 10.5cm, 12cm
- Number of strips: 2,4,6
- Angular width of each strip
- Gap between strips:
- Shape of strip
- Banking

Properties of Ball and Track

- Ball radius - 1.5cm
- Material of the track - Plastic
- surface roughness of the track
- Mass of the ball
- Coating - Aluminum foil
- Charge leakage or grounding conditions/ humidity

Voltage Variation



THANK YOU

-MALKOVA